JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)
WEEK 10: PROBABILITY

TOPIC: Definition of Probability; Importance and Usefulness; Examples of Chances/Events; Generating Events; Solving Problems; Calculating Probability

A. INTRODUCTION TO PROBABILITY

Definition:
Probability is the measure of how likely an event is to occur. It is the study of chance.

Key Terms:

Term	Definition	Example
1. Experiment	An action that produces a result	Tossing a coin, rolling a die, picking a card
2. Outcome	A possible result of an experiment	Heads or Tails (coin toss)
3. Event	A set of one or more outcomes	Getting an even number when rolling a die
4. Sample Space	All possible outcomes of an experiment	Sample space for coin toss = {Heads, Tails}

B. PROBABILITY SCALE

Range: Probability values range from 0 to 1

0 ---------- 0.5 ---------- 1
| | |

0 = Impossible (will never happen)
0.5 = Even chance (50-50)
1 = Certain (will definitely happen)

Examples on the Scale:

Probability	Type	Examples
0	Impossible	Sun rising in the west; Rolling a 7 on a standard die; Picking a red ball from a bag of only blue balls
0.5	Even chance	Tossing heads on a fair coin; Drawing a red or black card from a deck
1	Certain	Sun rising in the east; Picking a coloured ball from a bag with only coloured balls; Getting a number from 1-6
Between 0 and 1	Most real-life events	Raining tomorrow (maybe 0.3); Winning a game (maybe 0.7)

C. EXPRESSING PROBABILITY

Probability can be expressed as:

- Fraction: 1/2, 3/4, 2/5
- Decimal: 0.5, 0.75, 0.4
- Percentage: 50%, 75%, 40%

Conversions:

<u>Example 1:</u>

Fraction: 1/2

Decimal: $1 \div 2 = 0.5$

Percentage: $0.5 \times 100\% = 50\%$

<u>Example 2:</u>

Fraction: 3/4

Decimal: $3 \div 4 = 0.75$

Percentage: $0.75 \times 100\% = 75\%$

D. PROBABILITY FORMULA

<u>Basic Formula:</u>

P(Event) = Number of favourable outcomes / Total number of possible outcomes

OR:

$P(E) = n(E) / n(S)$

Where:

$P(E)$ = Probability of event E

n(E) = Number of outcomes in event E
n(S) = Total number of outcomes in sample space S

E. EXAMPLES WITH COINS

Experiment: Tossing One Coin
Sample Space: S = {H, T} where H = Heads, T = Tails
Total outcomes: n(S) = 2
Example 3: Probability of getting Heads
Favourable outcomes = 1 (just H)
Total outcomes = 2 (H or T)
P(Heads) = 1/2 = 0.5 = 50%
Example 4: Probability of getting Tails
P(Tails) = 1/2 = 0.5 = 50%

Experiment: Tossing Two Coins
Sample Space: S = {HH, HT, TH, TT}
Total outcomes: n(S) = 4

Example 5: Probability of getting 2 Heads
Favourable outcomes = 1 (HH only)
P(2 Heads) = 1/4 = 0.25 = 25%

Example 6: Probability of getting at least 1 Head
Favourable outcomes = 3 (HH, HT, TH)
P(at least 1 Head) = 3/4 = 0.75 = 75%

Example 7: Probability of getting exactly 1 Head
Favourable outcomes = 2 (HT, TH)
P(exactly 1 Head) = 2/4 = 1/2 = 50%

Example 8: Probability of getting no Heads (all Tails)
Favourable outcomes = 1 (TT)
P(no Heads) = 1/4 = 0.25 = 25%

F. EXAMPLES WITH DICE

Experiment: Rolling One Die
Sample Space: S = {1, 2, 3, 4, 5, 6}
Total outcomes: n(S) = 6

Example 9: Probability of rolling a 4
Favourable outcomes = 1 (just 4)
P(rolling 4) = 1/6 ≈ 0.167 ≈ 16.7%

Example 10: Probability of rolling an even number
Even numbers = {2, 4, 6}
Favourable outcomes = 3
P(even) = 3/6 = 1/2 = 0.5 = 50%

Example 11: Probability of rolling an odd number
Odd numbers = {1, 3, 5}
Favourable outcomes = 3
P(odd) = 3/6 = 1/2 = 50%

Example 12: Probability of rolling a number less than 4
Numbers less than 4 = {1, 2, 3}
Favourable outcomes = 3
P(less than 4) = 3/6 = 1/2 = 50%

Example 13: Probability of rolling a number greater than 4
Numbers greater than 4 = {5, 6}

Favourable outcomes = 2
P(greater than 4) = 2/6 = 1/3 ≈ 33.3%

Example 14: Probability of rolling a 7
No 7 on a standard die
Favourable outcomes = 0
P(rolling 7) = 0/6 = 0 (Impossible)

Example 15: Probability of rolling a number from 1 to 6
All outcomes are 1-6
Favourable outcomes = 6
P(1 to 6) = 6/6 = 1 (Certain)

<u>Experiment: Rolling Two Dice</u>
Sample Space: 36 possible outcomes

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

Example 16: Probability of getting a sum of 7

Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)
Number of favourable outcomes = 6
P(sum = 7) = 6/36 = 1/6 ≈ 16.7%

Example 17: Probability of getting doubles (same number on both dice)
Doubles: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6)

Favourable outcomes = 6
P(doubles) = 6/36 = 1/6 ≈ 16.7%

G. EXAMPLES WITH CARDS
Standard Deck: 52 cards
4 suits: Hearts (♥), Diamonds (♦), Clubs (♣), Spades (♠)
Each suit: 13 cards (Ace, 2-10, Jack, Queen, King)
Red cards: 26 (Hearts and Diamonds)
Black cards: 26 (Clubs and Spades)

Example 18: Probability of drawing a Heart
Number of Hearts = 13
Total cards = 52
P(Heart) = 13/52 = 1/4 = 0.25 = 25%

Example 19: Probability of drawing a red card
Red cards = 26
P(red) = 26/52 = 1/2 = 50%

Example 20: Probability of drawing a King
Number of Kings = 4
P(King) = 4/52 = 1/13 ≈ 7.7%

Example 21: Probability of drawing an Ace or a King
Aces = 4, Kings = 4
Total favourable = 8
P(Ace or King) = 8/52 = 2/13 ≈ 15.4%

H. PROBABILITY WITH COLOURS/BALLS
<u>Example 22:</u> A bag contains 5 red balls, 3 blue balls, and 2 green balls.

Total balls = 10

Question	**Calculation**	**Probability**
P(red) 5/10	1/2	= 50%
P(blue)3/10	3/10	= 30%
P(green)	2/10 = 1/5	= 20%
P(red or blue)	(5 + 3)/10 = 8/10 =4/5	= 80%
P(not green)	(10 - 2)/10 = 8/10=4/5	= 80%

<u>Example 23</u>: A box contains 12 marbles: 4 red, 5 blue, 3 yellow

Total = 12

Question	Calculation	Probability
P(red) 4/12	1/3 ≈ 33.3%	
P(blue)5/12	≈ 41.7%	
P(yellow)	3/12	1/4 = 25%
P(not yellow)	9/12	3/4 = 75%

I. **COMPLEMENTARY EVENTS**

<u>Definition</u>:

The complement of an event E is the event "not E", denoted as E' or Ē.

Rule:

P(E) + P(E') = 1

OR:

P(E') = 1 - P€

<u>Example 24:</u> If P(rain) = 0.3, find P(no rain)

P(no rain) = 1 - P(rain) = 1 - 0.3 = 0.7 = 70%

Example 25: If P(passing exam) = 4/5, find P(failing)

P(failing) = 1 - 4/5 = 5/5 - 4/5 = 1/5 = 20%

J. EXPERIMENTAL PROBABILITY

Definition:

Probability based on actual experiments or past observations.

Formula:

P(Event) = Number of times event occurred / Total number of trials

Example 26: A coin is tossed 50 times. Heads appears 28 times.

Experimental probability of Heads:

P(Heads) = 28/50 = 14/25 = 0.56 = 56%

Note: This differs from theoretical probability (50%), but approaches it with more trials.

Example 27: A die is rolled 60 times with results:

Number	**Frequency**
1	8
2	12
3	9
4	11
5	10
6	10

Experimental probabilities:

Number	Calculation	Probability
P(1)	8/60	2/15 ≈ 13.3%
P(2)	12/60	1/5 = 20%
P(3)	9/60	3/20 = 15%
P(4)	11/60	≈ 18.3%
P(5)	10/60	1/6 ≈ 16.7%
P(6)	10/60	1/6 ≈ 16.7%

Note: Theoretical probability for each number = 1/6 ≈ 16.7%

K. USING LUDO FOR PROBABILITY
Ludo Die: Same as standard die (1-6)
Activity: Roll the die multiple times and record results.

Example 28: Class experiment - Rolling a Ludo die
Each student rolls 10 times, class combines data:

Number	Frequency (out of 300 rolls)
1	48
2	52
3	49
4	51
5	50
6	50

Experimental probabilities: All close to 1/6 (theoretical) = 50/300 ≈ 16.7%
Observation: More trials → experimental probability gets closer to theoretical probability.

L. IMPORTANCE AND USEFULNESS OF PROBABILITY

Field	Application
1. Weather Forecasting	"70% chance of rain tomorrow" – helps people plan activities
2. Insurance	Calculate risk; determine premiums; life, health, car insurance based on probability
3. Medical Diagnosis	Probability of disease given symptoms; treatment success rates; drug effectiveness
4. Sports	Predicting match outcomes; player performance statistics; betting odds
5. Business Decisions	Risk assessment; investment decisions; market predictions; quality control
6. Games and Gambling	Casino games designed using probability; lottery odds; card game strategies
7. Agriculture	Crop yield predictions; weather impact on farming; pest control planning
8. Education	Test design; grading systems; predicting student success

9. Daily Life	Traffic planning; shopping decisions; time management
10. Science and Research	Hypothesis testing; experimental design; data analysis

M. PROBABILITY PROBLEMS

<u>**Example 29**</u>: A bag has 15 balls: 6 red, 5 blue, 4 green

Question	Calculation	Probability
● P(red)	6/15 2/5 =	0.4 = 40%
● P(not red)	1 - 2/5 = 3/5 OR (5+4)/15 = 9/15	3/5 = 60%
● P(blue or green)	(5 + 4)/15 = 9/15	3/5 = 60%

If 3 red balls are removed:
New: 3 red, 5 blue, 4 green = 12 total
P(red) = 3/12 = 1/4 = 25%

<u>**Example 30: Two coins are tossed**</u>

Question	Answer
1. List all possible outcomes	Sample space = {HH, HT, TH, TT}
2. P(2 heads)	1/4 = 25%
3. P(1 head and 1 tail)	Outcomes: HT, TH → 2/4 = 1/2 = 50%
4. P(at least 1 tail)	Outcomes: HT, TH, TT → 3/4 = 75%

M. <u>CLASSWORK</u>

Define probability.
What is sample space?
What probability value means "impossible"?
What probability value means "certain"?
A die is rolled. Find P(rolling 5).
A coin is tossed. Find P(Heads).
A bag has 3 red and 7 blue balls. Find P(red)
What is a complementary even

If P(event) = 0.6, find P(not event).
Give 3 real-life uses of probability.

O. REVISION QUESTIONS

- What is probability? Give 3 examples of experiments in probability. What is an outcome?
- What is sample space? What values can probability take? Express 3/4 as a decimal and percentage.
- State the probability formula. What is experimental probability? How many outcomes when tossing 2 coins?
- How many outcomes when rolling 1 die? In a deck of cards, how many red cards? What is complementary probability?
- Can probability be greater than 1? Why?, If an event is certain, its probability = ?, Name 5 fields that use probability.

P. ASSIGNMENT

SECTION A: Basic Concepts (15 marks)
1. Define:
a) Probability
b) Sample space
c) Event
d) Outcome

2. Classify each probability as: impossible (0), unlikely (0–0.5), even chance (0.5), likely (0.5–1), or certain (1)
a) 0.2
b) 1
c) 0.5
d) 0.9

e) 0

3. Convert:
a) 1/4 → decimal and percentage
b) 0.75 → fraction and percentage
c) 60% → fraction and decimal

SECTION B: Coin Tossing (20 marks)
1. One coin tossed. Find:
a) P(Heads)
b) P(Tails)

2. Two coins tossed. List sample space and find:
a) P(2 Heads)
b) P(2 Tails)
c) P(exactly 1 Head)
d) P(at least 1 Head)
e) P(no Heads)

3. Experiment: Toss a coin 30 times.

Record results

Calculate experimental P(Heads)

Compare with theoretical probability

SECTION C: Die Rolling (25 marks)
1. One die rolled. Find:
a) P(3)

b) P(even number)
c) P(odd number)
d) P(number > 4)
e) P(number ≤ 3)
f) P(7)
g) P(1, 2, 3, 4, 5, or 6)

2. Two dice rolled. Find:
a) Total possible outcomes
b) P(sum = 7)
c) P(doubles)
d) P(sum > 10)

3. Experiment: Roll a die 40 times.

Record frequency of each number

Calculate experimental probability for each

Compare with theoretical probability

SECTION D: Cards (15 marks)
A card is drawn from a standard 52-card deck. Find:

P(Heart)

P(red card)

P(black card)

P(King)

P(Ace)

P(face card – Jack, Queen, King)

P(number card 2–10)

P(red King)

P(not a Heart)

P(Ace or King)

SECTION E: Colours / Balls (20 marks)
1. Bag: 8 red, 5 blue, 7 green balls.
a) Total balls
b) P(red)
c) P(blue)
d) P(green)
e) P(not green)
f) P(red or blue)

2. Box: 15 marbles – 6 yellow, 4 white, 5 black.
a) P(yellow)
b) P(white)
c) P(black)
d) P(not black)
e) If 2 yellow marbles are removed, find new P(yellow)

3. Create your own experiment using coloured paper/objects.

Count different colours

Calculate probability for each colour

Draw one randomly 20 times

Calculate experimental probability

Compare results

SECTION F: Complementary Events (10 marks)
P(rain) = 0.4 → P(no rain) = ?

P(winning) = 3/5 → P(losing) = ?

P(passing) = 0.85 → P(failing) = ?

P(event A) = 2/7 → P(not A) = ?

P(sunny) = 60% → P(not sunny) = ?

SECTION G: Real-Life Application (15 marks)
Design a simple game using coins, dice, or coloured cards.

Write the rules

Calculate the probability of winning

Research and explain how probability is used in:

Weather forecasting

Sports

Your chosen career field

Explain why probability is important in decision-making.